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General Knowledge
Indian Geography

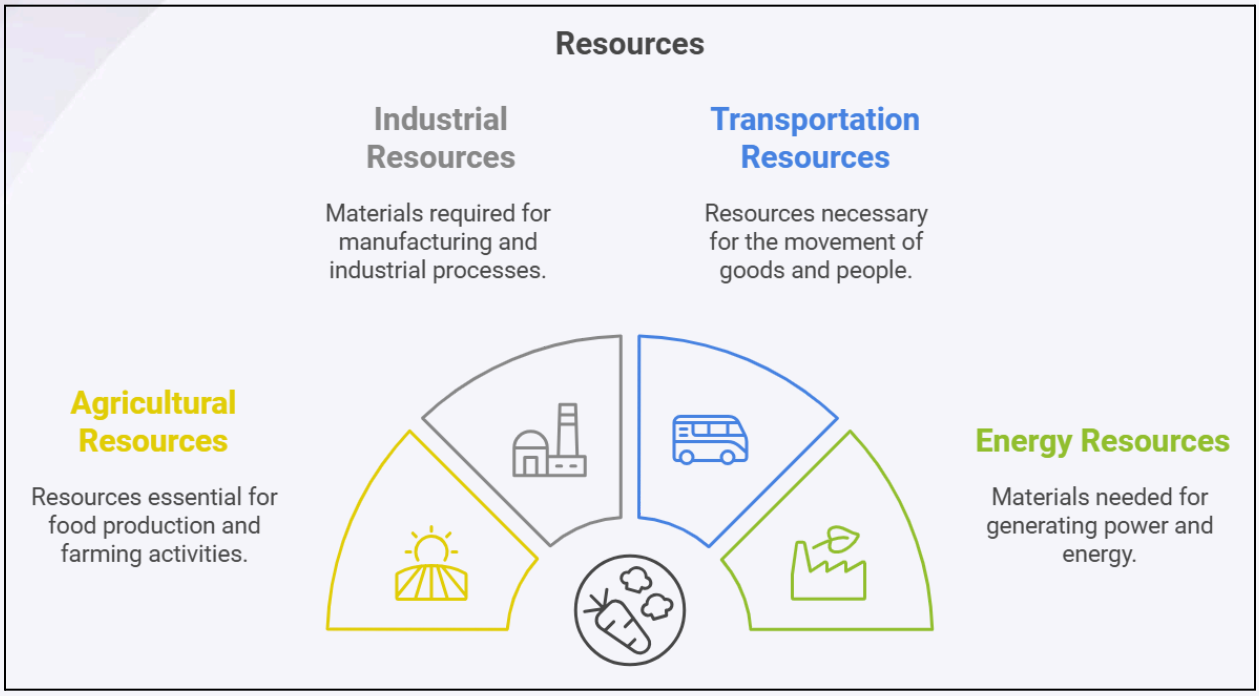
STUDY NOTES

Conventional & Non-Conventional Resources

Conventional and Non-Conventional Resources

Introduction

- Resources are materials naturally present in the environment that are essential for the survival and development of living organisms.
- They provide the foundational raw materials necessary for various human activities, including agriculture, industry, transportation, and energy production.
- These resources are critical for supporting life and fostering economic development.

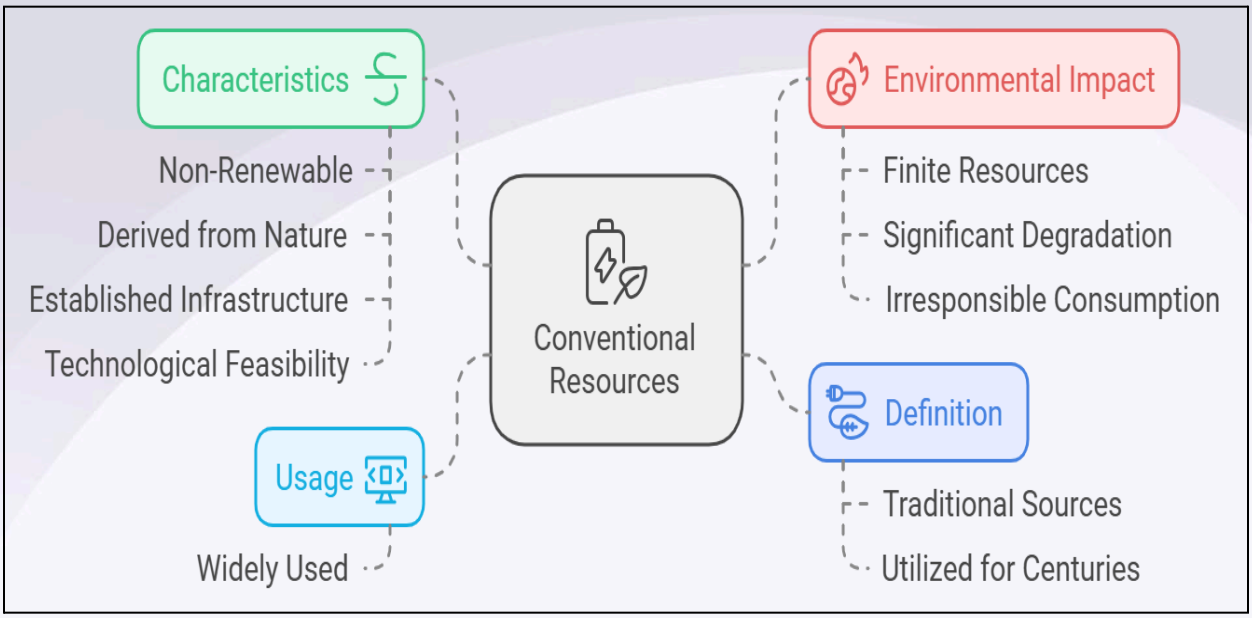


Based on their origin and usability, resources can be categorized into **two major types:** Conventional Resources and Non-Conventional Resources.

Conventional Resources

- **Conventional resources** refer to the traditional sources of energy and raw materials that have been utilized by humans for centuries.
- These resources are primarily **non-renewable** and are derived directly from nature.
- They are widely used because of their established infrastructure and technological feasibility.
- However, they are finite and can cause significant environmental degradation when consumed irresponsibly.

Characteristics of Conventional Resources



Examples of Conventional Resources

Coal:

- Used for generating electricity, powering industries, and running steam engines.
- Contributes significantly to air pollution and greenhouse gas emissions.

Petroleum:

- Known as crude oil, it is refined into various products such as petrol, diesel, kerosene, and cooking gas.
- Widely used in transportation, heating, and industrial operations.
- Found deep underground or beneath ocean floors.

Natural Gas:

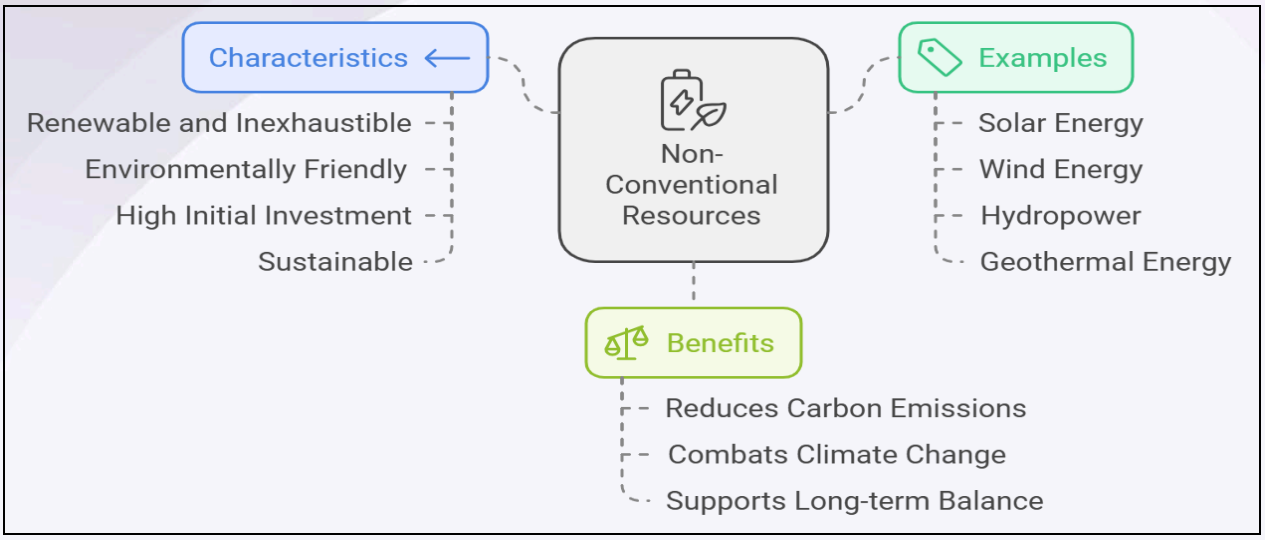
- A cleaner alternative to coal and petroleum, natural gas burns efficiently and emits less pollution.
- Used as Compressed Natural Gas (CNG) in vehicles and for generating electricity.

Nuclear Energy

- Derived from radioactive materials like uranium and thorium.
- Involves nuclear reactions that release heat energy, which is then used to produce electricity.

- While highly efficient, it carries risks of radioactive contamination and waste disposal issues.

Non-Conventional Resources



Examples of Non-Conventional Resources

Solar Energy

- Derived from the **sun's radiation** and converted into electricity or heat using solar panels or photovoltaic cells.
- Applications include **solar water heaters, solar cookers,** and large-scale solar power plants.
- Abundant, renewable, and entirely pollution-free.

Wind Energy

- Generated by harnessing the **kinetic energy** of wind through turbines.
- Wind farms, located in areas with high wind speeds, are used to produce electricity.
- A clean and sustainable energy source, though installation costs can be high.

Hydropower

- Energy is generated from the **movement of water**, typically from rivers or dams.
- Used to produce electricity through hydroelectric power plants.
- Reliable and efficient but may impact ecosystems and displace communities.

Biogas

- A renewable source of energy is produced through the **anaerobic decomposition of organic waste,** such as animal dung and plant residues.
- Widely used in rural areas for cooking and heating purposes.
- Reduces waste and greenhouse gas emissions.

Geothermal Energy

- Utilizes heat from **beneath the Earth's surface** to generate electricity and provide direct heating.
- A consistent and renewable energy source, though limited to geologically active regions.

Tidal Energy

- Generated by capturing the energy of **ocean tides using turbines.**
- Suitable for coastal areas with high tidal ranges.
- Renewable and predictable but expensive to harness.

Conventional vs. Non-Conventional Resources

Indicator	Conventional Resources	Non-Conventional Resources
Definition	Traditional energy sources like coal, oil, and gas.	Modern, eco-friendly options like solar and wind energy.
Renewability	Non-renewable; takes millions of years to form.	Renewable; replenishes naturally and won't run out.
Availability	Limited and found in specific regions.	Abundant and available globally.
Examples	Coal, petroleum, natural gas, nuclear energy.	Solar, wind, hydropower, geothermal, biogas.
Environmental Impact	High pollution and greenhouse gas emissions.	Clean energy with little to no pollution.
Sustainability	Unsustainable due to depletion and damage to nature.	Highly sustainable for long-term use.

Energy Efficiency	Moderate; energy loss during production.	High; efficient energy conversion.
Carbon Emissions	A significant contributor to global warming.	Minimal or zero emissions.
Technology	Uses old, well-established tech.	Relies on advanced, evolving tech.
Examples of Use	Power plants, vehicles, industries.	Homes, industries, rural energy, clean electricity.